

Problem 12.1 A horizontal cantilever of uniform section of length l carries two point loads, W at the free end and $2W$ at a distance of a from the free end. Find the maximum deflection due to this loading. If the cantilever is a steel tube of circular section 100 mm external diameter and 6 mm thick and $l=1.50$ meters and $a = 0.60$ meters determine the value of W so that the maximum bending stress is 140 N/mm^2 and calculate the maximum deflection for the loading. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$.

Solution: Deflection at the free end due to the load W alone

$$= \frac{Wl^3}{3EI}$$

Deflection at the free end due to the load $2W$ alone

$$= \frac{2W(l-a)^3}{3EI} + \frac{2W(l-a)^2}{2EI}a$$

$$\therefore \text{Total deflection at the free end } y = \frac{Wl^3}{3EI} + \frac{2W(l-a)^3}{3EI} + \frac{W(l-a)^2}{EI}a$$

$$y = \frac{W}{3EI} [l^3 + 2(l-a)^3 + 3(l-a)^2 a]$$

$$= \frac{W}{3EI} [l^3 + 2l^3 - 6l^2a + 6la^2 - 2a^3 + 3l^2a - 6la^2 + 3a^3]$$

$$= \frac{W}{3EI} [3l^3 - 3l^2a + a^3]$$

$$\therefore y = \frac{W}{3EI} [3l^2(l-a) + a^3]$$

For the numerical

$$\text{Max B.M. } M = Wl + 2W(l-a)$$

$$W(3l-2a) = W(3 \times 1.5 - 2 \times 0.6) \text{ Nm} = 3.3 W \text{ Nm.}$$

$$I = \frac{\pi}{64} (100^4 - 88^4) \text{ mm}^4 = 1.965 \times 10^6 \text{ mm}^4$$

$$\text{Stress due to max. B.M. } = \sigma = \frac{M}{I} y = 140 \text{ N/mm}^2$$

$$\therefore \frac{3.3W \times 1000}{1.965 \times 10^6} \times 50 = 140$$

$$\therefore W = \frac{140 \times 1.965 \times 10^6}{3.3 \times 1000 \times 50} \text{ N} = 1667 \text{ N.}$$

$$\text{Max deflection } y = \frac{W}{3EI} [3l^2(l-a) + a^3]$$

$$= \frac{1667}{3 \times 2.1 \times 10^5 \times 1.965 \times 10^6} [3 \times 2.25 \times 0.9 + 0.216] \times 1000^3 \text{ mm}$$

$$= 8.47 \text{ mm.}$$

Problem 12.2 A cantilever of length 2 meters carries a uniformly distributed load of 5000 n per meter for a length of 1.25 meter from the fixed end and a point load of 1000 N at the free end. If the section is rectangular 120 mm side and 240 mm deep, find the deflection at the free end. Take $E = 10000 \text{ N/mm}^2$.

Solution: Moment of inertia of the section

$$= I = \frac{bd^3}{12}$$

$$= \frac{120 \times 240^3}{12}$$

$$= 1.3824 \times 10^8 \text{ mm}^4$$

Deflection at C due to uniformly distributed load on AB

$$= \frac{wa^4}{8EI} + \frac{wa^3}{6EI}(l-a)$$

$$= \frac{5000 \times 1.25^4}{8EI} + \frac{5000 \times 1.25^3 \times 0.75}{6EI} = \frac{2146}{EI}$$

Deflection at C due to point load at C

$$= \frac{Wl^3}{3EI} = \frac{2000 \times 2^3}{3EI} = \frac{5334}{EI}$$

$\therefore$ Net deflection at C

$$\frac{2746}{EI} + \frac{5334}{EI} = \frac{8080}{EI} = \frac{8080(1000)^3}{10^4 \times 1.3824 \times 10^8} \text{ mm}$$

$$= 5.844 \text{ mm}$$

Problem: 12.3 A cantilever beam 120 mm wide and 20 mm deep is 2.5 m long. What uniformly distributed load should the beam carry to produce a deflection of 5 mm at the free end and what concentrated load at free end would produce the same deflection ? give the slope at the free end in both cases. Ignore self-weight of the beam. Take $E = 2 \times 10^5 \text{ N/mm}^2$.

Solution: Let I = moment of inertia of the rectangular section about centroidal axis =

$$\frac{1}{12} \times 120 \times 200^3 = 8000 \times 10^4 \text{ mm}^4$$

Maximum deflection at the free end of the cantilever

$$\frac{wL^4}{8EI} = 5$$

$$\frac{w \times (2500)^4}{8 \times 2 \times 10^5 \times 8 \times 10^7} = 5$$

$$\therefore w = 16.384 \text{ N/mm}$$

$$\text{Slope at the free end} = \frac{wL^3}{6EI} = \frac{16.384 \times 2500^3}{6 \times 2 \times 10^5 \times 8 \times 10^7} = 2.666 \times 10^{-3} \text{ rad.}$$

Let W Newton be the concentrated load that can be placed under the given conditions of the problem.

$$\text{Maximum deflection at the free end} = \frac{wL^3}{3EI} = 5$$

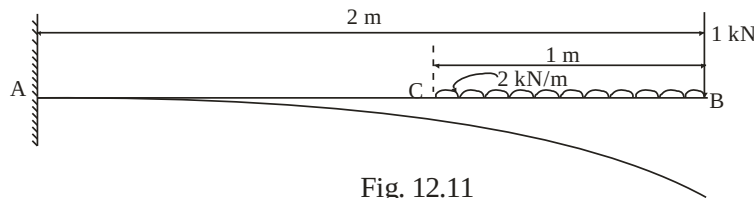
$$\text{or } \frac{W \times 2500^3}{3 \times 2 \times 10^5 \times 8 \times 10^7} = 5$$

$$W = 15360 \text{ N} = 15.36 \text{ kN}$$

$$\text{Slope at the free end} = \frac{wL^2}{2EI} = \frac{W \times 2500^2}{2 \times 2 \times 10^5 \times 8 \times 10^7} = 0.003 \text{ rad.}$$

Problem 12.4 A cantilever 100 mm wide and 200 mm deep projects 2 m from a wall into which it is cast. The cantilever carries a u.d.l. of 2 kN/m over a length of 1 m from the free end and a point load of 1 kN at the free end. Find the slope and deflection at the free end. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$.

Solution: Here, as in Fig. 12.11



$$L = AB = 2\text{m}$$

$$L_1 = AC = 1\text{m}$$

$$W = 1 \text{ kN}, w = 2 \text{ kN/m}$$

Moment of inertia of the beam section,

$$I = \frac{1}{12} \times 100 \times (200)^3 = \frac{2}{3} \times 10^8 \text{ mm}^4.$$

Slope at the free end B is given by

$$\begin{aligned} i_B &= \frac{WL^2}{2EI} + \left[\frac{wL^3}{6EI} - \frac{w(L-L_1)^3}{6EI} \right] = \frac{1}{EI} \left[\frac{WL^2}{2} + \frac{wL^3}{6} - \frac{w(L-L_1)^3}{6} \right] \\ &= \frac{1}{2.1 \times 10^5 \times \frac{2}{3} \times 10^8} \times \left[\frac{2 \times 10^3 \times 2000^2}{2} + \frac{2 \times 2000^3}{6} - \frac{2 \times (2-1)^3 \times 1000^3}{6} \right] \end{aligned}$$

$$= 0.000309 \text{ radian}$$

$$W = 10^3 \text{ N}$$

$$L = 2000 \text{ mm} \quad L_1 = 1000 \text{ mm}$$

$$w = 2 \text{ N/mm}$$

Deflection at the free end B is given by

$$\begin{aligned} y_B &= \frac{WL^3}{3EI} + \frac{wL^4}{8EI} - \left[\frac{w(L-L_1)}{8EI} + \frac{w(L-L_1)^3 \times L_1}{6EI} \right] \\ &= \frac{10^3 \times 2^3 \times 3 \times 10^4}{3 \times 210 \times 10^9 \times 2} + \frac{2 \times 10^3 \times 2^4 \times 3 \times 10^4}{8 \times 210 \times 10^9 \times 2} \\ &\quad - \left[\frac{2 \times 10^3 \times (2-1)^4 \times 3 \times 10^4}{8 \times 210 \times 10^9 \times 2} + \frac{2 \times 10^3 \times (2-1)^3 \times 3 \times 10^4 \times 1}{6 \times 210 \times 10^9 \times 2} \right] \\ &= 0.00045 \text{ m} = 0.45 \text{ mm},. \end{aligned}$$

Problem 12.5 A simply supported beam is loaded as shown in Fig 12.12. Determine the deflection at points C and D in the beam. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 2 \times 10^8 \text{ mm}^4$.

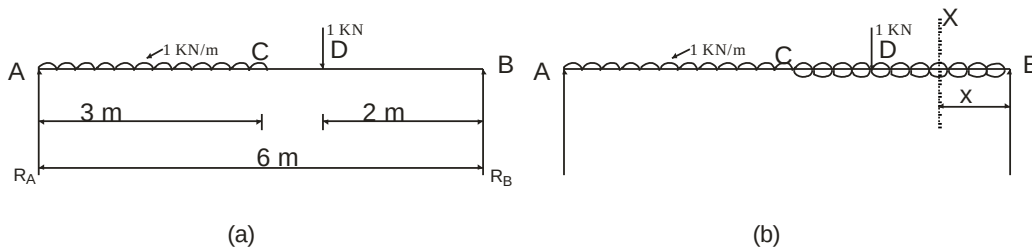


Fig.12.12

Solution: Taking moment about B

$$R_A \times 6 = 1000 \times 3 \times 4.5 + 1000 \times 2$$

$$\therefore R_A = \frac{15500}{6} \text{ N}$$

On portion BC a u.d.l. of 1 kN/m is introduced both above and below so that these loads neutralise each other and the net effect remains unchanged. Consider a section X-X at a distance x from left end A.

$$M = \frac{15500x}{6} - \frac{1000x^2}{2} - 1000(x-4) + \frac{1000}{2}(x-3)^2$$

$$EI \frac{d^2 y}{dx^2} = -M = -\frac{15500x}{6} + \frac{1000x^2}{2} + 1000(x-4) + \frac{1000}{2}(x-3)^2$$

Integrating the above equation for slope, we get

$$EI \frac{dy}{dx} = -\frac{15500x^2}{12} + \frac{1000x^3}{6} + \frac{1000(x-4)^2}{2} + \frac{1000(x-3)^3}{6} + C_1$$

Integrating again for deflection, we have

$$EI.y = -\frac{15500x^3}{36} + \frac{1000x^4}{24} + \frac{1000(x-4)^3}{6} + \frac{1000(x-3)^4}{24} + C_1x + C_2$$

$$\text{At } x = 0, \text{ deflection is zero, i.e. } y = 0 \quad \therefore C_2 = 0$$

$$\text{At } x = 6, \text{ deflection is zero, i.e. } y = 0 \quad \therefore C_1 = 6.84 \times 10^3$$

Therefore the final equation for slope is

$$EI \frac{dy}{dx} = -\frac{15500x^2}{12} + \frac{1000x^3}{6} + \frac{1000(x-4)^2}{6} + \frac{1000(x-3)^4}{24} + 6.84 \times 10^3$$

The final equation for deflection is

$$EI.y = -\frac{15500x^3}{36} + \frac{1000x^4}{24} + \frac{1000(x-4)^3}{6} + \frac{1000(x-3)^4}{24} + 6.84 \times 10^3 x$$

For deflection at C put $x = 3$

$$y_c = \frac{1}{EI} \left[-\frac{15500 \times 3^3}{36} + \frac{1000 \times 3^4}{24} + 6.84 \times 10^3 \times 3 \right] = \frac{12.27 \times 10^3}{EI}$$

$$= \frac{12.27 \times 10^3}{2 \times 10^5 \times 2 \times 10^8} = 0.3 \text{ mm}$$

For deflection at D put $x = 4$

$$y_D = \frac{1}{EI} \left[-\frac{15500 \times 4^3}{36} + \frac{1000 \times 4^4}{24} - \frac{1000}{24} + 6.84 \times 10^3 \times 4 \right]$$

$$= \frac{10.429 \times 10^3}{EI} = \frac{10.429 \times 10^3}{2 \times 10^5 \times 2 \times 10^8} = 0.26 \text{ mm.}$$

Problem 12.6 A rolled steel beam having a span of 6 meters carries a point load of 80 kN at 4 meters from the left support. Find the deflection under the load and the position and amount of maximum deflection.

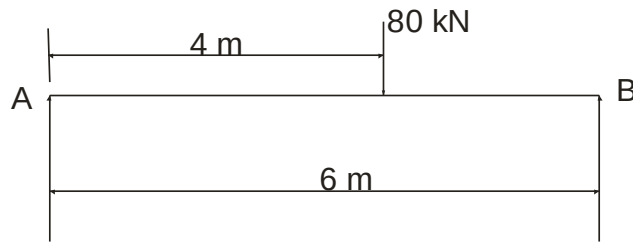


Fig.12.13

I_{xx} for the section = $7.33 \times 10^7 \text{ mm}^4$. Take $E = 200 \text{ kN/mm}^2$.

Solution: Fig. 12.13 shows the beam carrying the given loading.

Bigger segment = $AC = a = 4 \text{ m}$.

Smaller segment = $BC = b = 2 \text{ m}$.

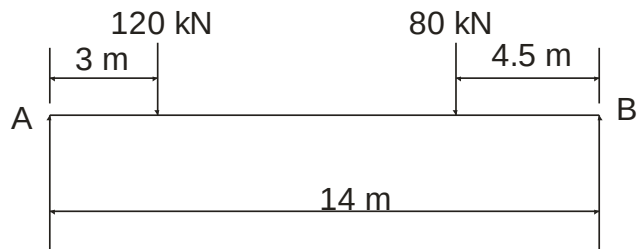
$$\begin{aligned} \text{Deflection under the load} &= y_c = \frac{Wa^2b^2}{3EI l} \\ &= \frac{80 \times 4^2 \times 2^2}{3 \times 200 \times 7.33 \times 10^7 \times 6} (1000)^3 \text{ mm} = 19.4 \text{ mm} \end{aligned}$$

Maximum deflection occurs at

$$\sqrt{\frac{l^2 - b^2}{3}} = \sqrt{\frac{36 - 4}{3}} = 3.27 \text{ meters from the left end.}$$

$$\begin{aligned} \text{Max. deflection} &= y_{\max} = \frac{Wb(l^2 - b^2)^{3/2}}{9\sqrt{3} EI l} \\ &= \frac{80 \times 2(32)^{3/2} (1000)^3}{9\sqrt{3} (200 \times 7.33 \times 10^7 \times 6)} = 21.12 \text{ mm} \end{aligned}$$

Problem 12.7 A horizontal girder of steel having uniform section is 14 meters long and is simply supported at its ends. It carries concentrated loads of 120 kN and 80 kN at two points 3 meters and 4.5 meters from the two ends respectively. I for the section of the girder is $32 \times 10^8 \text{ mm}^4$. and $E_s = 210 \text{ kN/mm}^2$. Calculate the deflections of the girder at points under the two loads.



Find also the maximum deflection.

Solution: Let R_a and R_b be the vertical reactions at the supports A and B.

Taking moments about A, we have

$$R_b \times 14 = 120 \times 3 + 80 \times 9.5$$

$$\therefore R_b = 80 \text{ kN} \quad R_a = 200 - 80 = 120 \text{ kN}$$

The B.M. at any section distance x from A is given by,

$$EI \frac{d^2 y}{dx^2} = 120x - 120(x-3) - 80(x-9.5)$$

Integrating, we get,

$$EI \frac{dy}{dx} = 60x^2 + C_1 - 60(x-3)^2 - 40(x-9.5)^2$$

$$\text{Integrating again, we get} \quad EIy = 20x^3 + C_1x + C_2 - 20(x-3)^3 - \frac{40}{3}(x-9.5)^3$$

$$\text{At } x = 0$$

$$y = 0$$

$$\therefore C_2 = 0$$

$$\text{At } x = 14,$$

$$y = 0$$

$$\therefore 0 = 20(14)^3 + 14C_1 - 20(14-3)^3 - \frac{40}{3}(14-9.5)^3$$

$$\therefore 14C_1 = -27045 \quad \therefore C_1 = -1931.78$$

Hence the deflection at any section is given by

$$EIy = 20x^3 - 1931.78x - 20(x-3)^3 - \frac{40}{3}(x-9.5)^3$$

To find the deflection at C, put $x = 3\text{m}$ in the deflection equation

$$EIy_c = 20 \times 3^3 - 1931.78 \times 3 = -5255.34$$

$$y_c = -\frac{5255.34}{210 \times 32 \times 10^8} (1000)^3 = -7.82 \text{ mm}$$

$$y_c = 7.82 \text{ mm (downward equation)}$$

to find the deflection at D, put $x = 9.5 \text{ m}$ in the deflection equation

$$EIy_d = 20(9.5)^3 - 1931.78 \times 9.5 - 20(9.5-3)^3$$

$$EIy_d = -6696.91$$

$$y_d = - \frac{6696.91}{210 \times 32 \times 10^8} (1000)^3 \text{ mm} = - 9.965 \text{ mm}$$

$$y_d = 9.965 \text{ (downward deflection)}$$

Maximum deflection: Let us assume that the deflection will be maximum at a section between C and D.

Equating the slope at the section to zero, we have,

$$EI \frac{dy}{dx} = 60x^2 - 1931.78 - 60(x - 3)^2 = 0$$

Solving, we get $x = 6.87 \text{ m.}$

Substituting in the deflection equation, we get,

$$EI y_{\max} = 20(6.87)^3 - 1931.78 \times 6.87 - 20(6.87 - 3)^3$$

$$EI y_{\max} = - 7945.7$$

$$y_{\max} = - \frac{7945.7}{210 \times 32 \times 10^8} \times (1000)^3 \text{ mm} = - 23.6 \text{ mm}$$

i.e. $y_{\max} = 11.8 \text{ mm (downward deflection)}$

Problem 12.8 For the loaded beam shown in Fig. 12.15 find (i) the slopes at A, C and D (ii) the deflection at C and D and (iii) the maximum deflection. Take $E = 200 \text{ kN/mm}^2$ and $I = 9.25 \times 10^7 \text{ mm}^4$.

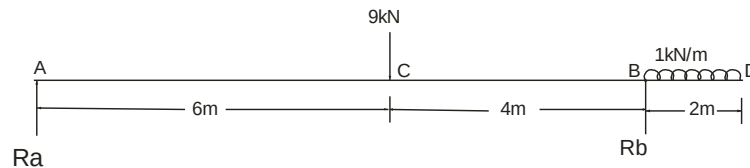


Fig. 12.15

Solution: Taking moments about A, $R_b \times 10 = (9 \times 6) + (1 \times 2 \times 11) \quad \therefore R_b = 7.60 \text{ kN}$

$$\therefore R_a = 11 - 7.60 = 3.40 \text{ kN}$$

The equation of the deflected shape of the beam is given by,

$$EI \frac{d^2 y}{dx^2} = \text{B.M.}$$

$$EI \frac{d^2 y}{dx^2} = 3.40x - 9(x - 6) + 7.60(x - 10) - \frac{1}{2}(x - 10)^2$$

Integrating again, $EIy = \frac{1.70}{3}x^3 + C_1x + C_2 - 1.50(x-6)^3 + (x-10)^3 - (x-10)^4$

At $x = 0, y = 0 \therefore C_2 = 0$

At $x = 10, y = 0$

$$\therefore 0 = \frac{1.70}{3} \times 10^3 + 10C_1 - 1.50 \times 4^3 \therefore C_1 = -47.067$$

Hence, the slope and deflection equations are,

$$EI \frac{dy}{dx} = 1.70x^2 - 47.067x - 4.50(x-6)^2 + 3.80(x-10)^2 - \frac{1}{6}(x-10)^3 \text{ .. (slope equation)}$$

$$EIy = \frac{1.70}{3}x^3 - 47.067x - 1.50(x-6)^3 + \frac{3.80}{3}(x-10)^3 - \frac{1}{24}(x-10)^4 \dots \text{ (deflection equation)}$$

Slope at A: Put $x = 0$ in the slope equation,

$$EIi_a = -47.067 \quad i_a = \frac{-47.067}{200 \times 9.25 \times 10^7} \times 1000^2 = -0.002544 \text{ radian}$$

Slope at C: Put $x = 6$ m in the slope equation,

$$EIi_c = 1.70 \times 6^2 - 47.067 = 14.133$$

$$i_c = \frac{14.133}{200 \times 9.25 \times 10^7} \times 1000^2 = 0.0007639 \text{ radian}$$

Slope at D: Put $x = 12$ m in the slope equation,

$$EIi_d = 1.70 \times 12^2 - 47.067 - 4.50 \times 6^2 + 3.80 \times 2^2 - \frac{1}{6} \times 2^3 = 49.60$$

$$i_d = \frac{49.60}{200 \times 9.25 \times 10^7} \times 1000^2 = 0.002681 \text{ radian}$$

Deflection at C: Put $x = 6$ m in the deflection equation,

$$EIy_c = \frac{1.70}{3} \times 6^3 - 47.067 \times 6 - 160.002 \therefore y_c = - \frac{160.002}{200 \times 9.25 \times 10^7} \times 1000^3 = -8.65 \text{ mm}$$

Deflection at D: Put $x = 12$ m in the deflection equation.

$$\therefore y_d = \frac{99.863}{200 \times 9.25 \times 10^7} \times 1000^3 = 5.4 \text{ mm}$$

Maximum deflection

Assuming the deflection to be maximum at a section between general expressions for slope to zero,

$$0 = 1.70 x^2 - 47.067 \quad \therefore x^2 = \frac{47.067}{1.7}$$

$$\therefore x = 5.262 \text{ m}$$

(Since this distance is less than 6m, our assumption about the maximum deflection is correct). Putting $x = 5.262 \text{ m}$ in the deflection.

$$EIy_{\max} = \frac{1.70}{3} \times 5.262^3 - 47.067 \times 5.26$$

$$\therefore y_{\max} = - \frac{165.105}{200 \times 9.25 \times 10^7} \times 1000^3 = -8.925 \text{ mm}$$

Problem 12.9 Fig. 12.16 (a) shows a simply supported beam of uniform section whose moment of inertia is $4.3 \times 10^8 \text{ mm}^4$. For the loading shown, find the position and magnitude of the maximum deflection. Take $E = 210 \text{ kN/mm}^2$.

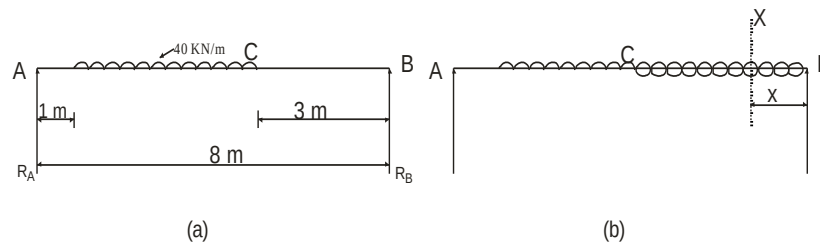


Fig.12.16

Solution: Taking moments about the end A, we have

$$R_b \times 8 = 40 \times 4 \times 3$$

$$R_b = 6 \text{ kN}$$

$$R_a = 40 \times 4 - 60 \text{ kN} = 100 \text{ kN}$$

In order the general expression for the bending moment at any section at any section may be expressed in the form suitable for application of Macaulay's Method the loading on the beam is arranged as shown in Fig. 12.16 (b).

Now, following Macaulay's method, the B.M. at any section distant x from A is given by,

$$EI \frac{d^2 y}{dx^2} = 100x \left| -\frac{40(x-1)^2}{2} \right| + \frac{40(x-5)^2}{2}$$

$$EI \frac{d^2 y}{dx^2} = 100x \left| -20(x-1)^2 \right| + 20(x-5)^2$$

Integrating, we get, $EI \frac{dy}{dx} = 50x^2 + C_1 \left| -\frac{20}{3}(x-1)^3 \right| + \frac{20}{3}(x-5)^3$

Integrating again, we get $EIy = \frac{50}{3}x^3 + C_1x + C_2 \left| -\frac{5}{3}(x-1)^4 \right| + \frac{5}{2}(x-5)^4$

At A, the deflection is zero

At $x = 0, y = 0$

$\therefore C_2 = 0$

At B also the deflection is zero,

$\therefore$ At $x = 8, y = 0$

$\therefore 0 = \frac{50 \times 8^3}{3} + 8C_1 - \frac{5 \times 7^4}{3} + \frac{5 \times 3^4}{3}$

$\therefore C_1 = -583.33$

Position of maximum deflection

Let us assume that the deflection will be maximum between C and D.

Equating the slope to zero, we have,

$$50x - 583.33 - \frac{20}{3}(x-1)^3 = 0$$

or $15x^2 - 175 = 2(x-1)^3$

solving the above equation by trial and error we get $x = 3.82$ meters.

The value of x obtained confirm that our assumption about the position of maximum deflection is correct.

Substituting $x = 3.82$ meters in the deflection equation, we get

$$EIy_{\max} = \frac{50}{3} \times 3.82^3 - 583.33 \times 3.82 - \frac{5}{3} \times 2.82^4 = -1404.67$$

$\therefore y_{\max} = -\frac{1404.67}{210 \times 4.3 \times 10^6} \times (1000)^3 \text{ mm} = -15.55 \text{ mm}$

Problem 12.10 A beam has supports l apart with equal overhangs $\frac{l}{3}$ over each support. It carries a load W at each end and a load $2W$ at the centre. Find the slopes at the free end and at the support. Find also the deflections at the free end and at the center.

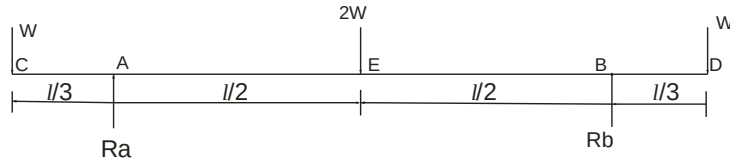


Fig. 12.17

Solution: Fig. 12.17 shows the beam subjected to the given load system.

By symmetry, $R_a = R_b = 2W$

We will consider one half of the beam length.

At any section distance x from the free end C, the bending moment is given by,

$$EI \frac{d^2 y}{dx^2} = -Wx \left| + 2W \left(x - \frac{l}{3} \right) \right.$$

Integrating,
$$EI \frac{dy}{dx} = -\frac{Wx^2}{2} + C_1 \left| + W \left(x - \frac{l}{3} \right)^2 \right.$$

$$\text{At } x = \frac{5l}{6}, \frac{dy}{dx} = 0$$

$$\therefore 0 = -\frac{25}{72}Wl^2 + C_1 + \frac{Wl^2}{4} \quad \therefore C_1 = \frac{7}{72}Wl^2$$

$$\therefore EI \frac{dy}{dx} = -\frac{Wx^2}{2} + \frac{7}{72}Wl^2 \left| + W \left(x - \frac{l}{3} \right)^2 \right. \quad \dots (\text{Slope equation})$$

Integrating again,
$$EI y = -\frac{Wx^3}{6} + \frac{7}{72}Wl^2 x + C_2 \left| + \frac{W}{3} \left(x - \frac{l}{3} \right)^3 \right.$$

$$\text{At } x = \frac{l}{3}, y = 0$$

$$\therefore 0 = -\frac{Wl^3}{162} + \frac{7}{216}Wl^3 + C_2 \quad \therefore C_2 = -\frac{17}{648}Wl^3$$

$$\therefore EIy = -\frac{Wx^3}{6} + \frac{7}{72}Wl^2x - \frac{17}{648}Wl^3 \left| + \frac{W}{3}\left(x - \frac{l}{3}\right)^3 \right. \text{ (deflection equation)}$$

Slope at C: Putting $x = 0$ in the slope equation,

$$EI i_c = \frac{7}{72}Wl^2 \quad \therefore i_c = \frac{7}{72} \frac{Wl^2}{EI}$$

Slope at A: Putting $x = \frac{l}{3}$ in the slope equation,

$$EI i_a = -\frac{Wl^2}{18} + \frac{7}{72}Wl^2 = \frac{Wl^2}{24}$$

$$\therefore i_a = \frac{Wl^2}{24EI}$$

Deflection at C: Putting $x = 0$ in the deflection equation,

$$EI y_c = -\frac{17}{648}Wl^3 \quad \therefore y_c = -\frac{17}{648} \frac{Wl^3}{EI}$$

Deflection at E: Putting $x = \frac{5l}{6}$ in the deflection equation,

$$\begin{aligned} EI y_e &= -\frac{W}{6}\left(\frac{5l}{6}\right)^3 + \frac{7}{72}Wl^2 \frac{5l}{6} - \frac{17}{648}Wl^3 + \frac{W}{3} \frac{l^3}{8} \\ &= \left[-\frac{125}{1296} + \frac{35}{432} - \frac{17}{468} + \frac{1}{25} \right] Wl^3 = 0 \end{aligned}$$

$$\therefore y_e = 0$$

Alternative Solution:

The above problem can also be solved easily by the principal of super-position.

First load system : Loads W each at the ends C and D only

Second load system ; Load $2W$ at the centre only.

Analysis due to the first load system

In this case it may be noted that the part of the beam AB between the supports is subjected to a constant bending moment $M = \frac{Wl}{3}$ Hence due to this load system,

$$\text{Slope at A} = \frac{ML}{2EI} = \frac{Wl}{3} \cdot \frac{l}{2EI} = \frac{Wl^2}{6EI} \text{ (anticlockwise)}$$

Slope at C = Slope at A + Slope at C consideration AC as a cantilever fixed at A

$$\frac{Wl^2}{6EI} + W \left(\frac{l}{3} \right)^2 \frac{1}{2EI} = \frac{Wl^2}{6EI} + \frac{Wl^2}{18EI} = \frac{2}{9} \cdot \frac{Wl^2}{EI} \text{ (anticlockwise)}$$

Deflection at C = Deflection at C due to slope at A + Deflection at C considering AC as a cantilever fixed at A

$$= \frac{Wl^2}{6EI} \cdot \frac{l}{3} + \frac{W}{3EI} \left(\frac{l}{3} \right)^3 = \frac{Wl^3}{18EI} + \frac{Wl^2}{81EI} = \frac{11}{162} \frac{Wl^3}{EI} \downarrow$$

$$\text{Deflection at the center} = \frac{Wl^2}{8EI} = \frac{Wl}{3} \cdot \frac{l^2}{8EI} = \frac{Wl^3}{24EI} \uparrow$$

Analysis due to the second load system

In this case,

$$\text{Slope at A} = \text{Slope at C} = \frac{2W \cdot l^2}{16EI} = \frac{Wl^2}{8EI} \text{ (clockwise)}$$

$$\text{Deflection at C} = \frac{Wl^2}{8EI} \times \frac{l}{3} = \frac{Wl^3}{24EI} \uparrow$$

$$\text{Deflection at the center} = \frac{2Wl^3}{48EI} = \frac{Wl^3}{24EI} \downarrow$$

Hence,

$$\text{Net slope at A} = \frac{Wl^2}{6EI} - \frac{Wl^2}{8EI} = \frac{Wl^2}{24EI} \text{ (anticlockwise)}$$

$$\text{Net slope at C} = \frac{2}{9} \frac{Wl^2}{EI} - \frac{Wl^2}{8EI} = \frac{7}{72} \frac{Wl^2}{EI} \text{ (anticlockwise)}$$

$$\text{Net deflection at the centre} = \frac{Wl^3}{24EI} - \frac{Wl^3}{24EI} = 0$$

Exercise

12.1 Derive expression for the slope and deflection of a beam subjected to uniform bending moment.

12.2 What is Macaulay's method? Where is it used? Find an expression for the deflection of a simply supported beam which is subjected to eccentric point load using this method.

12.3 What are the different methods of finding slope and deflection of a cantilever?

12.4 A wooden beam 4 m long simply supported at its ends carries a point load of 7.25 kN at its centre. The cross section of the beam is 140 mm wide and 240 mm deep. If E for the beam is $6 \times 10^3 \text{ N/mm}^2$, find the deflection at the centre.

(Ans. 10 mm)

12.5 A beam of length 10 m is simply supported at its ends carries two point loads of 100 kN and 60 kN at a distance of 2 m and 5 m respectively from the left support. Calculate the deflection under each load. Find also the maximum deflection. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 18 \times 10^8 \text{ mm}^4$.

(Ans. -4.35, -6.76, -6.78 mm)

12.6 A beam of length 20 m is simply supported at its ends carries two point loads of 4 kN and 10 kN at a distance of 8 m and 12 m respectively from the left support. Calculate the deflection under each load. Find also the maximum deflection. Take $E = 2 \times 10^6 \text{ N/mm}^2$ and $I = 1 \times 10^9 \text{ mm}^4$.

(Ans. -10.3, -10.6, -11 mm)

12.7 A cantilever beam of length 3 m carries a point load of 60 kN at a distance of 2 m from the fixed end. Calculate the slope and deflection at the free end. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 1 \times 10^8 \text{ mm}^4$.

(Ans. 0.006 radian, 14 mm)

12.8 A cantilever beam of length 3 m carries a udl over the entire length. If the slope at the free end is 0.01777 radians calculate the deflection at the free end.

(Ans. 39.99 mm)

12.9 A cantilever beam of length 4 m carries a udl 12 kN/m over a length of 3 m from the fixed end. Calculate the slope and deflection at the free end. Take $EI = 2 \times 10^{13} \text{ Nmm}^2$.

(Ans. 0.0027 radian, 8.775 mm)

12.10 A beam of length 4.8 m and of rectangular section is simply supported at its ends carries a udl of 9.375 kN/m over the entire length. Calculate the width and depth of the beam if permissible bending stress is 7 N/mm^2 and the maximum deflection should exceed 9.5 mm. Take $E = 1.05 \times 10^4 \text{ N/mm}^2$.

(Ans. 240, 336.8 mm)